I/O Architecture

The peripheral devices are the main vehicle for obtaining the benefits from the system since they are responsible for the input/output (I/O) and for connecting the system to the outside world. Some devices have a dual role, such as mass storage, networks, and so on—these are both I/O devices.

In general, these peripheral devices can be divided into the following:

- **Input peripherals** – This allows user input from the outside world to the computer.
- **Output peripherals** – This allows information output from the computer to the outside world.
- **Input-output peripherals** – This allows information to be sent through input and output.
- **Storage** – This allows the usage of I/O for storing and fetching of information.

I/O Characteristics

These are the following characteristics to consider in dealing with I/O devices and controllers:

- Computer usefulness ultimately depends on its I/O capabilities.
- I/O devices are incredibly diverse with respect to the following:
  - **Behavior** – input, output, or storage purposes
  - **Partner** – human or machine interaction
  - **Data rate** – the amount of data transferred to or from the I/O device in a period. It is typically measured in *bits per second*
- I/O considerations:
  - **Performance** – This refers to how fast the device is.
  - **Expandability** – This refers to the expansion of the I/O device.
  - **Dependability** – This refers to the capability of the range and control of an I/O device.
  - **Cost** – This refers to the cost (how cheap or expensive) of an I/O module or device.
  - **Size and Weight** – These refer to the dimensions of an I/O device.

Device	Input/Output	Data Rate*	Control
Keyboard	Input	Very Low	External & Software
Mouse	Input	Low	External
Touchpad			
Touch Screen			
Scanner	Input	Medium	External & Software
Microphone	Input	Low to Medium	External & Software
Sound	Input/Output	Medium	Program
USB	Input/Output	Low to Very High	External & Software
Network	Input/Output	High to Very High	External & Software
Printer	Output	Low to Medium	Program
Graphics Display	Output	High	Program
Flash Drive	Storage	Medium	External & Software
Magnetic Disk	Storage	Medium	Program
Solid State Drive	Storage	Medium to High	Program
Optical Drive	Storage	Medium to High	External & Software
Magnetic Drive	Storage	Low to Medium	External & Software

Table 1. Characteristics of Typical I/O Devices

***Very Low** < 500 bps; **Low** < 5 Kbps; **Medium** < 10 Mbps; **High** 10 – 100 Mbps; **Very High** 100 – 5000 Mbps

I/O Performance

Below are the parameters used in checking the performance of each I/O device or controller.

- **I/O bandwidth (throughput)** – This is the amount of information that can be input (output) and communicated across an interconnect (say, for example, a bus) to the processor or memory (I/O device) per unit time.
    - How much data can be moved through the system at a certain time?
    - How many I/O operations can be done per unit time?
- **I/O response time (latency)** – This is the total elapsed time to accomplish by an input or output operation. This is also an especially important performance metric in real-time systems.

Modes of I/O Data Transfer

Data transfer between the central unit and I/O devices can be handled in three (3) types of modes:

- **Programmed I/O** – These instructions are the result of I/O instructions written in a computer program. The instruction in the program initiates each data item transfer. Usually, the program controls data transfer to and from the CPU and peripheral. Transferring data under programmed I/O requires constant monitoring of the peripherals by the CPU.

  The CPU has the following pair of registers to interface with an I/O device:
    - **Data register** for holding I/O data
    - **Address register** for addressing the I/O device.
- **Interrupt Initiated I/O** – The interface determines when the peripheral is ready for data transfer, then it generates an interrupt. After receiving the interrupt signal, the CPU stops the task it processes and service the I/O transfer. Then, it returns to its previous processing task.

  Interrupts require CPU immediate attention (e.g., keyboard input, power failure, completion of I/O, etc.). These are the following uses of interrupts:
    - **As an external event notifier.** This process periodically checks for external events (ex. checking keyboard input).
    - **As a completion signal.** It can be a notifier when a process is complete.
    - **As a means of allocating CPU time.** The time between interrupt pulses is known as a *quantum*, which represents the time that each program or process will have allotted to it.
    - **The interrupt as an abnormal event indicator.** It uses traps or exceptions to avoid errors or illegal instructions.
    - **Software interrupts.** Programs or applications with higher priority and privileges (depending on the user's and system's grant) can request a special interrupt if prompted.
- **Direct Memory Access (DMA)** – It is a technique of removing the CPU from the path and letting the peripheral device manage the memory buses directly to improve the speed of transfer. A DMA controller manages the transfer of data between peripherals and memory unit. Many hardware systems use DMA, such as disk drive controllers, graphic cards, network cards, sound cards, etc.

  In DMA, the CPU initiates the transfer, does other operations while the transfer is in progress, and receives an interrupt from the DMA controller when the transfer has been completed.

  The three (3) main conditions to meet for a DMA to work are as follows:
    - Method to connect the I/O module and the memory
    - The I/O module must be able to read from and write to the memory
    - Method to avoid conflict between the CPU and the I/O module writing to memory at the same time.

There are different DMA methods of execution:

- **Buffer Chaining**
  - It handles multiple transfers without the processor.
  - The device gives a linked list of buffers.
  - The device hardware uses the next buffer on the list automatically.
- **Operation Chaining**
  - Further optimization for smart device
  - The processor gives a series of commands to the device, sometimes called a *channel program*.
  - The device carries out successive commands automatically

I/O Controller

The I/O controller serves as an intermediary interface between the CPU and a specific device that accepts commands from the CPU and controls the exact device after. Most of the I/O controllers are device controllers that serve as direct interfaces between a general system bus and each of the system's peripheral devices. In general, I/O controllers simplify the task of interfacing peripheral devices to a CPU.

I/O controllers offload a considerable amount of work from the CPU. They make it possible to control I/O to a peripheral with a few simple I/O commands from the CPU. They also support DMA so the CPU may be free to perform other tasks. Device controllers provide the specialized circuitry required to interface different types of peripherals to the computer. I/O controllers allow the processing of each instruction to progress in parallel.

These are the following functions of an I/O controller:

- **Interface translation** – It includes the connection, voltage supply, protocol enactor, clocking.
- **Addressing** – It is able to process memory locations or addresses for the function of processing.
- **Multiplexing** – It can combine multiple signals over the bus to reduce multiple bus usage.
- **Buffering** – It gives data transfer a boost by preloading data into memory before processing.
- **Error detection and correction** – It can detect errors and correction.
- **Control of multiple steps** – This reduces CPU workload due to I/O controller being the one that processes I/O instructions.

Computer Drivers

A device driver is a small piece of software that tells the operating system and other software how to communicate with a piece of hardware. Device drivers are like translators between a program being used and a device that program wants to utilize somehow. In other words, a software program can provide information to a driver to explain what it wants a piece of hardware to do, information the device driver understands and can fulfill with the hardware.

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